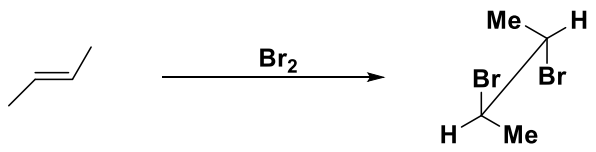
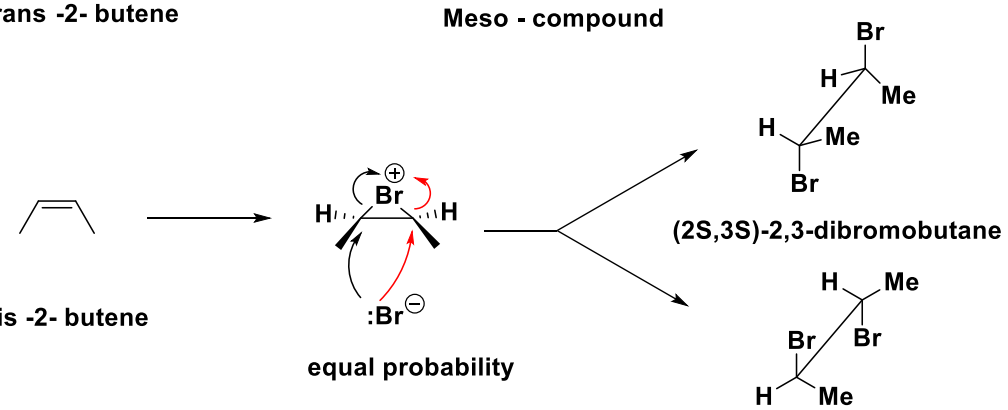
Class 2

- **The bromination reaction is anti - stereoselective and stereospecific**



Trans -2- butene

Meso - compound



Cis -2- butene

equal probability

(2S,3S)-2,3-dibromobutane

(2R,3R)- 2,3-dibromobutane

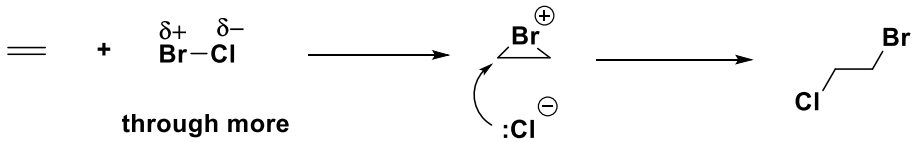
formed as racemic mixture

- **Reactions of interhalogen compounds and halogens in different solvents**

  - **Inter halogen compounds**

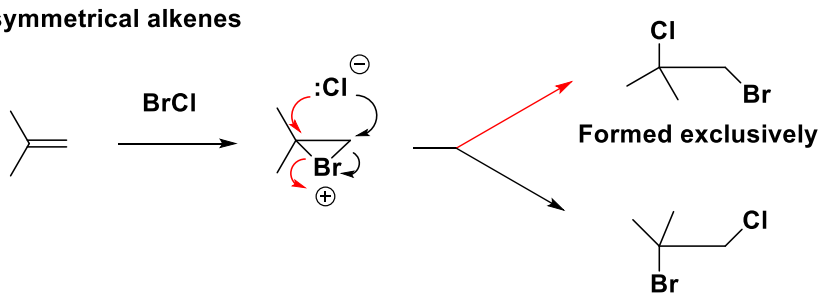
reactivity order : BrCl > Br_2 > ICl >IBr > I_2

For symmetrical alkenes



through more positively polarised end

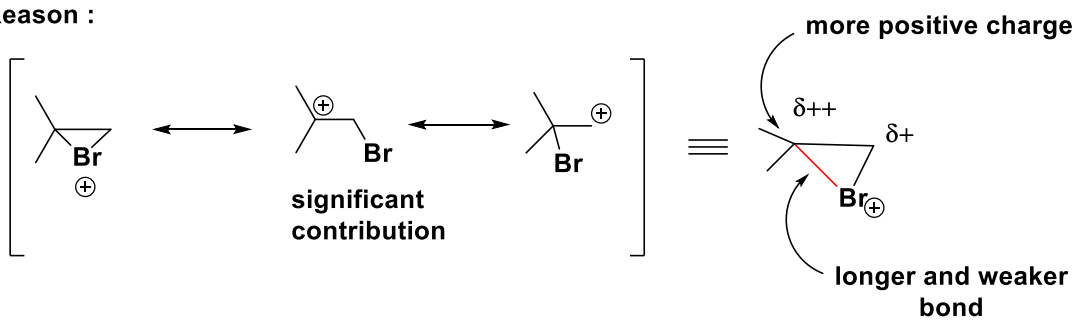
For unsymmetrical alkenes



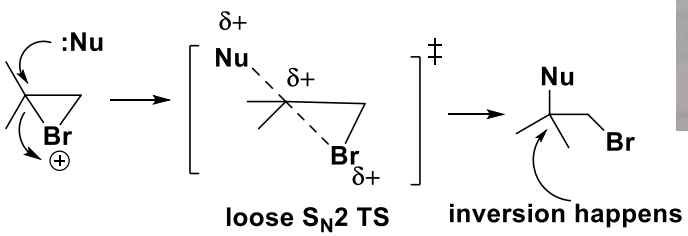
Formed exclusively

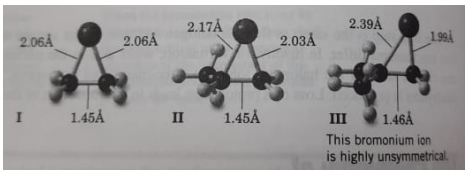
Rule : attack at the more substituted or more positive charge stabilizing carbon centre (valid for the opening of any 3- membered ring with a positive charge)

Reason :

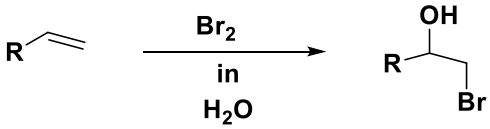


reaction proceeded via loose S_N2 TS

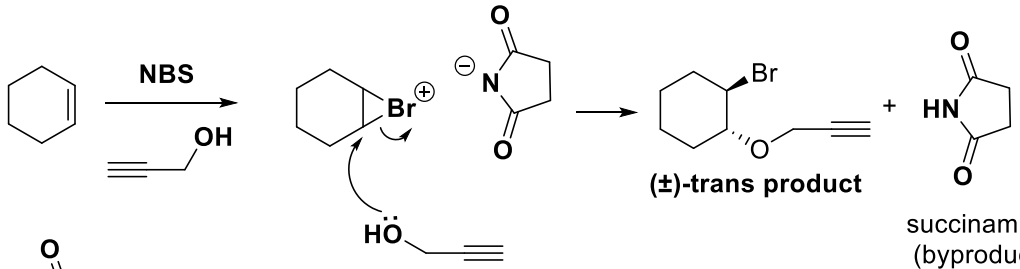


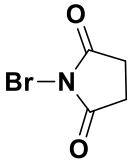


Same concept for
Br_2 in H_2O and Br_2 in MeOH



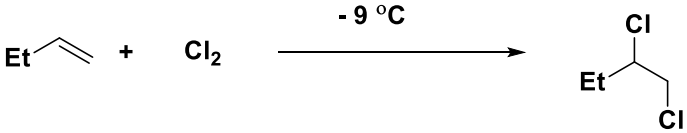
NBS = N-bromo succinamide can be used for better selectivity



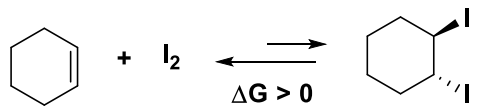


= NBS

Similar chlorination reaction is known and proceed via same mechanism



Similar iodination reaction is reversible



$\Delta H = \Delta H$ (**bond broken**) **+** ΔH (**bond formed**)

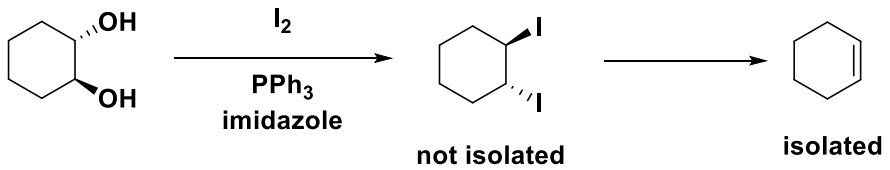
$= \Delta H\ (\pi_{C=C} + \sigma_{I-I}) + \Delta H\ (2 * \sigma_{C-I})$

$= +\ (64+3) - (2*51)$

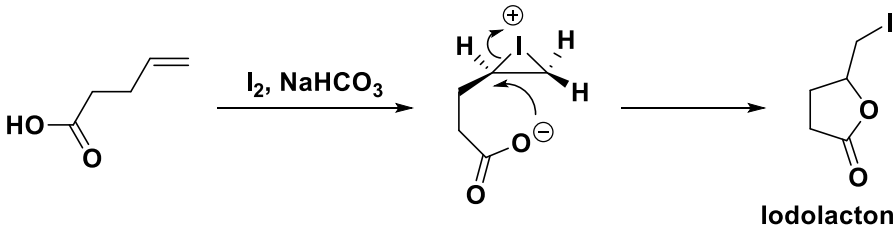
$= -2$ Kcal/ mol **(not sufficient to overcome the entropic penalty)**

$\Delta G = \Delta H - T\Delta S > 0$

Evidence :



Iodolactonization reaction :

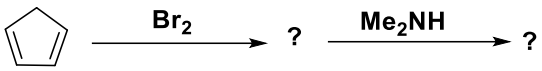


Iodolactone
(versatile synthetic intermediate)

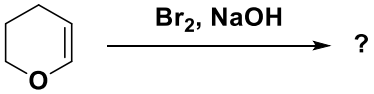
similar bromolactonization is also known

Questions:

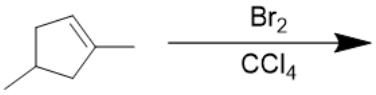
1. When cyclohexene is allowed to react with bromine in an aqueous solution of sodium chloride, the products are trans-1,2-dibromocyclohexane, trans-2-bromocyclohexanol and trans- 1-bromo-2-chlorocyclohexane. Write a plausible mechanism that explain the formation of these products.
2. Predict the product with stereochemistry.

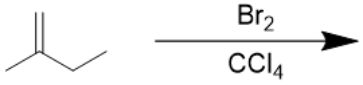


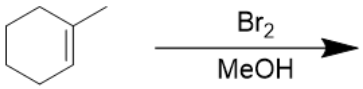
3. The major product formed in the following reaction is:



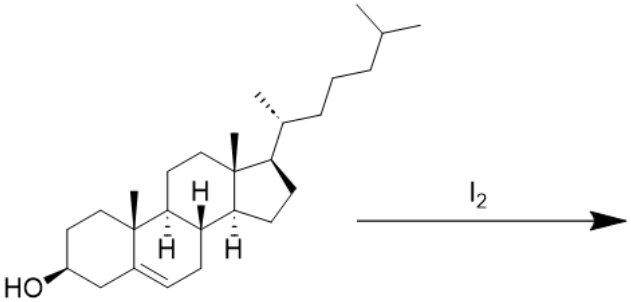
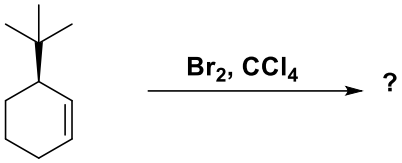
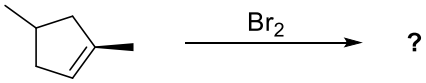
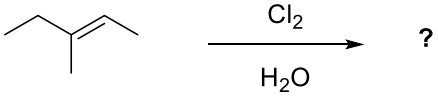
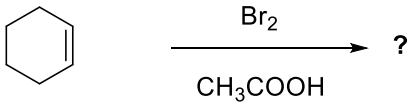
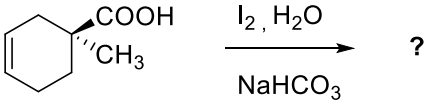
4. Write all possible products (major and minor) of the following reactions with mechanism, also state the relation between these products.

a. 

b. 

c. 

5. State whether the reaction will happen or not, if yes, write the product, if not, mention reason.



6. Predict the product with stereochemistry.